



TB AUDIO PLUG-INS
Team Impulse Impact

USER MANUAL

TB Compressor

Peak / RMS Compressor with Saturator

Precise peak or RMS compression with lookahead, an external sidechain with filters, and an optional saturator with circuit-derived colour models.

Peak / RMS detector

Soft knee

External sidechain

6 saturation types · 5 models

Up to 8x oversampling



Contents

- | | | | |
|-----------|-------------------------------|-----------|--------------------------|
| 01 | Overview | 07 | Saturator |
| 02 | Quick start | 08 | Presets |
| 03 | Interface | 09 | Recipes |
| 04 | Dynamics controls | 10 | Tips and troubleshooting |
| 05 | Gain, performance, and output | 11 | Parameter reference |
| 06 | Sidechain | 12 | Version history |
| | | 13 | Support and license |

01 Overview

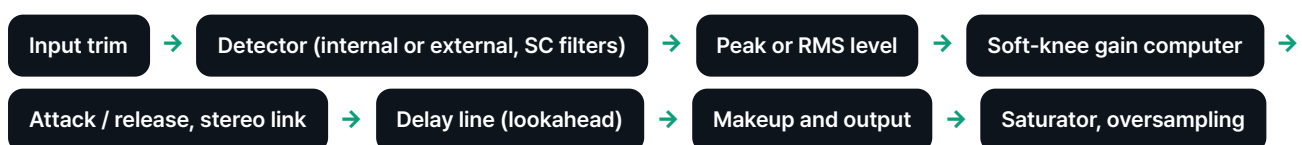
TB Compressor is a single-band compressor for tracks, buses, and full mixes. It reduces the level of loud passages above a threshold, and lets you shape how quickly it reacts, how smoothly it starts, and what it listens to.

Next to the classic controls you get a peak or RMS detector, lookahead, stereo linking, an external sidechain input with its own filters, and a saturator that adds harmonic colour after compression. Oversampling up to 8x keeps the saturator clean.

Key features

- Threshold, ratio, soft knee, attack, release, and makeup gain
- Peak detection, or 50 ms RMS detection that follows the average level
- Up to 20 ms lookahead so the compressor can react before a peak arrives
- External sidechain with high-pass and low-pass filters on the detector
- Stereo link, so both channels receive the same gain reduction
- Saturator with six types (Clean, Soft, Hard, Punch, Warm, Tape) and five circuit-derived models
- 1x, 2x, 4x, or 8x oversampling
- Transfer curve, six-second level history, and IN, OUT, and GR readouts

Signal flow



About this manual

This manual describes TB Compressor 2.1.0 for Windows (VST3). The current macOS build is version 1.1.3 and may differ from the descriptions here.

02 Quick start

- 1 Insert TB Compressor on a track or bus. The defaults are a gentle starting point: **THRESH** -18 dB, **RATIO** 2:1, **KNEE** 6 dB, **ATTACK** 10 ms, **RELEASE** 100 ms.
- 2 Play the loudest part and lower **THRESH** until the GR readout **3** shows the amount of reduction you want, typically 2 to 6 dB.
- 3 Set **ATTACK**: longer times let the start of each note through for more punch; shorter times catch peaks.
- 4 Set **RELEASE** so the gain recovers in time with the music, without audible pumping.
- 5 Raise **MAKEUP** **8** until the compressed signal matches the bypassed level, then compare with bypass.
- 6 Optional: switch on the **SATURATOR** **11** and raise **DRIVE** for colour, or connect an external sidechain **10** for ducking.

03 Interface



TB Compressor interface. The sidechain filters appear when an external sidechain is active.

- 1 **Preset browser** — previous and next preset, and the current Factory or user preset.
- 2 **Save, Undo, Redo** — save the current state to a .tbcpreset file and step through recent changes.
- 3 **IN, OUT, GR** — input and output level and the current gain reduction.
- 4 **CURVE** — the transfer curve: input level against output level, with the current ratio.
- 5 **Level history** — the last six seconds of level, with the threshold (THR) as a dashed line.
- 6 **DYNAMICS** — **THRESH**, **RATIO**, **KNEE**, **ATTACK**, and **RELEASE**.
- 7 **Detector** — **PEAK** or **RMS**.
- 8 **GAIN** — **INPUT** trim before the detector, and **MAKEUP** after compression.
- 9 **PERFORMANCE** — oversampling factor and **LOOKAHD** (lookahead).
- 10 **SIDECHAIN** — shows whether the internal (INT) or an external (EXT) key is used. With EXT, the SC HPF and SC LPF filters become active.

11 SATURATOR — on/off, **DRIVE**, **MIX**, the circuit model, and the saturation type.

12 OUTPUT — final **OUTPUT** level and **ST LINK** (stereo link).

04 Dynamics controls

Control	Range · default	What it does
THRESH	−60 to 0 dB · −18 dB	Compression threshold. Signals above it are reduced; the lower it is, the more of the program is compressed.
RATIO	1:1 to 20:1 · 2:1	Compression ratio: how strongly level above the threshold is reduced. High ratios limit transients firmly.
KNEE	0 to 6 dB · 6 dB	Knee width. A wide knee starts compressing gently just below the threshold; 0 dB gives a hard knee.
ATTACK	0.1 to 200 ms · 10 ms	How quickly gain reduction reaches its target. Too fast dulls the attack of notes; too slow lets the first peak through.
RELEASE	5 to 1000 ms · 100 ms	How quickly the gain recovers. Short times can pump; long times can keep the next note down.
Detector (PEAK / RMS)	Peak, RMS · Peak	Peak reacts to instantaneous peaks. RMS follows a 50 ms average and controls sustained energy rather than short peaks.

05 Gain, performance, and output

Control	Range · default	What it does
INPUT	−24 to +24 dB · 0 dB	Input trim before the detector and delay line. It changes how hard the compressor is driven.
MAKEUP	0 to 24 dB · 0 dB	Gain added after compression to restore level.
OUTPUT	−24 to +24 dB · 0 dB	Final output level.
LOOKAHD	0 to 20 ms · 0 ms	Lookahead: lets the detector react before a peak reaches the output. It adds latency.
Oversampling	1x (Off), 2x, 4x, 8x · 1x	Runs the processing at a higher sample rate to reduce aliasing from saturation. Higher settings use more CPU and add latency.
ST LINK	Off, On · On	Links gain reduction across channels so the stereo image stays steady. Off lets each channel compress on its own.

TB Compressor uses a delay line of up to 20 ms for lookahead, and oversampling adds further latency. It is designed for mixing rather than zero-latency monitoring while recording.

06 Sidechain

By default the detector listens to the audio being compressed (INT). When your DAW sends a signal to TB Compressor's sidechain input, the indicator changes to EXT and the detector listens to that key signal instead, for example a kick drum that ducks a bass or a voice that ducks music. The external key is used only when the sidechain bus is active in your DAW.

Control	Range · default	What it does
SC HPF	20 Hz to 10 kHz · 20 Hz	High-pass filter on the external key (host name Sidechain Low Cut). Removes low frequencies from what the detector hears.
SC LPF	100 Hz to 20 kHz · 20 kHz	Low-pass filter on the external key (host name Sidechain High Cut). Removes high frequencies from what the detector hears.

The sidechain filters shape only what the detector hears. They do not filter the audio you hear, and they work only with an external key (FILTERS ON WITH EXT SC).

07 Saturator

The saturator sits after compression and adds harmonic colour. Switch it on with **ON**, set the amount with **DRIVE**, and blend it with **MIX**. The circuit model selects the character: Legacy, or the circuit-derived VCA, opto, FET, and vari-mu colour models XXL, LX-2X, FET-X76, and FX-67X. The type selector sets the Legacy saturation type.

Control	Range · default	What it does
ON	Off, On · Off	Enables the saturator.
DRIVE	0 to 4 · 1	Saturation drive. Higher values add more harmonics and density.
MIX	0 to 1 · 1	Dry/wet blend of the saturator.
Circuit model	Legacy, XXL, LX-2X, FET-X76, FX-67X · Legacy	Saturator circuit model.
Type	Clean, Soft, Hard, Punch, Warm, Tape · Clean	Legacy saturation type.

Compare the saturator at matched loudness. Saturation adds harmonics and level, which can make it seem better than it is.

08 Presets

Step through presets with the arrows in the preset browser **1**. LOAD FROM FILE opens a saved .tbcpreset file, and Save stores the current state as a file. Undo and Redo step through recent changes.

Factory presets: **Clean Control**, **Vocal Leveler**, **Drum Punch**, **Bus Glue**, and **Sidechain Duck**.

09 Recipes

Starting points for common jobs. Adjust them by ear and check the GR readout.

Level a vocal

Even out a vocal performance without squashing it.

Preset	Vocal Leveler
RATIO	2:1 to 4:1
ATTACK	5 to 15 ms
RELEASE	60 to 150 ms
GR	3 to 6 dB on loud phrases

- 1 Lower THRESH until only the loud phrases are reduced.
- 2 Match the level with MAKEUP.

Add punch to drums

More impact from kick, snare, or a drum bus.

Preset	Drum Punch
ATTACK	10 to 30 ms
RELEASE	50 to 100 ms
RATIO	4:1

- 1 Lengthen ATTACK until the hit's initial crack comes through.
- 2 Shorten RELEASE until the gain recovers before the next hit.

Glue a mix bus

Make a mix feel more cohesive.

Preset	Bus Glue
RATIO	1.5:1 to 2:1
KNEE	6 dB
GR	1 to 3 dB
ST LINK	On

- 1 Keep reduction small; the bus should breathe, not pump.

Duck bass with the kick

Make room for the kick by ducking the bass on each hit.

Preset	Sidechain Duck
Key	Kick to the sidechain input (EXT)
SC LPF	Around 150 Hz to focus on the kick
ATTACK	Short
RELEASE	Timed to the groove

- 1 Route the kick to TB Compressor's sidechain input in your DAW.
- 2 Lower THRESH until the bass dips on each kick.

10 Tips and troubleshooting

Symptom	What to check
The track sounds better only because it is louder.	Match levels with MAKEUP or OUTPUT and compare with bypass at the same loudness.
Drums lose their snap.	Lengthen ATTACK, lower RATIO, or try the PEAK detector with less reduction.
The level pumps audibly.	Lengthen RELEASE or reduce the amount of gain reduction.
The stereo image shifts while compressing.	Switch ST LINK on.
The sidechain does not react.	Check that your DAW routes a signal to the sidechain input and that the indicator shows EXT.
The SC filters do nothing.	They affect only an external key, and only what the detector hears.

Symptom	What to check
The plug-in adds latency.	Lookahead and oversampling add latency. Set LOOKAHD to 0 ms and oversampling to 1x for the lowest delay.

11 Parameter reference

Panel label	Host parameter	Range	Default
INPUT	Input	−24 to +24 dB	0 dB
THRESH	Threshold	−60 to 0 dB	−18 dB
RATIO	Ratio	1:1 to 20:1	2:1
ATTACK	Attack	0.1 to 200 ms	10 ms
RELEASE	Release	5 to 1000 ms	100 ms
KNEE	Knee	0 to 6 dB	6 dB
MAKEUP	Makeup	0 to 24 dB	0 dB
OUTPUT	Output	−24 to +24 dB	0 dB
LOOKAHD	Lookahead	0 to 20 ms	0 ms
Detector	RMS	Off (peak), On (RMS)	Off
ST LINK	Stereo Link	Off, On	On
SC HPF	Sidechain Low Cut	20 Hz to 10 kHz	20 Hz
SC LPF	Sidechain High Cut	100 Hz to 20 kHz	20 kHz
Oversampling	Oversampling	1x (Off), 2x, 4x, 8x	1x
Saturator ON	Saturation Enabled	Off, On	Off
DRIVE	Saturation Drive	0 to 4	1
MIX	Saturation Mix	0 to 1	1
Type	Saturation	Clean, Soft, Hard, Punch, Warm, Tape	Clean
Circuit model	Saturator Circuit Model	Legacy, XXL, LX-2X, FET-X76, FX-67X	Legacy

12 Version history

Version	Changes
2.1.0 · 2026-08-09	Clearer input and gain-reduction readouts on the main display. Refined sidechain state feedback and meter visibility; the 2.0 control layout is unchanged.

13 Support and license

Report problems or request features at github.com/TaebeumKim/TB_Audio-Plugins/issues. Include the plug-in version, your DAW, and your operating system.

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